

netflow: a generic sparse network solver, demonstrated across thermal, hydraulic, and neutronic physics

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Abstract

We present **netflow**, a Python framework that solves problems reducible to *conservation of a flux at network nodes*. A small, domain-agnostic core (nodes carrying a scalar state, edges relating two states to a conserved flux, Newton/Picard solves over `scipy.sparse` [5]) hosts physics as plugins. We show the core is generic not merely across *domains* (thermal, hydraulic, neutronic) but across *problem structures* (driven boundary-value, eigenvalue, transient, coupled multiphysics)—all expressed as an *outer loop of driven core solves*, with no core modifications. Verification is exact against textbook closed forms. In a *code-to-code* comparison against published VERA Core Physics Benchmark solutions [1, 2] (whose Problems 6/7 have no reference solution), netflow’s peak *volume-average* fuel temperature agrees with the MC21/CTF solution to within 6%, and its calibrated subchannel exit-coolant spread brackets the published code spread. A Doppler-coupled neutronics/thermal solve computes the power self-consistently, exhibiting the negative temperature feedback that makes reactors stable.

1 Motivation and abstraction

Many engineering domains share one mathematical skeleton: a quantity is conserved at the nodes of a network, and each edge carries a flux that depends on its two endpoint states. Heat conduction (Fourier), pipe flow (Darcy–Weisbach), neutron diffusion (Fick), and electrical circuits (Ohm) are all instances. **netflow** factors this skeleton into a core with three objects: a **Node** (scalar state, optional Dirichlet/Neumann boundary, optional capacitance), an **Edge** ($\text{flux}(s_a, s_b)$) and an optional analytic Jacobian), and a **Network** that assembles the residual

$$F_i(\mathbf{x}) = \text{source}_i + \sum_{e=(a,b)} [\mathcal{K}_{b=i} f_e - \mathcal{K}_{a=i} f_e] = 0 \quad (1)$$

and solves it with damped Newton (analytic sparse Jacobian) or a Picard fallback. The core has *zero* domain imports; a CI “layering” test enforces it. Figure 1 shows the same abstraction instantiated in four domains.

2 The meta-abstraction: outer loops

The core solves *driven* problems $F(\mathbf{x}) = 0$. Three larger problem classes reduce to repeated driven solves (Fig. 2, right):

Genericity Rosetta stone — one core, many domains, three problem structures

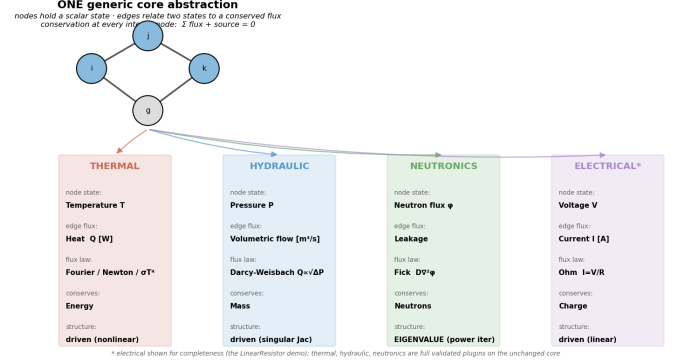


Figure 1: One core abstraction across four domains and three problem structures. State, flux, flux-law, conserved quantity, and mathematical structure differ; the network solver does not.

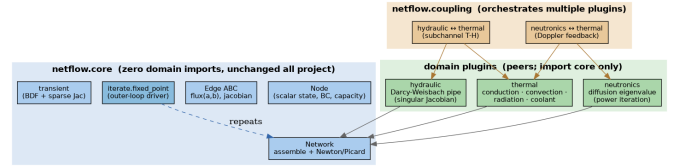


Figure 2: Layered architecture. The core (blue, unchanged for the entire project) is wrapped by peer domain plugins (green) and a coupling layer (orange) that orchestrates several plugins. Plugins never import each other; coupling code may.

- **Transient** ($C_i \dot{T}_i = F_i$): implicit BDF time stepping [5]; each step is a driven Newton solve. A naïve dense Jacobian costs $\mathcal{O}(N^2)$ memory (15 GB at a full fuel assembly); supplying the analytic sparse Jacobian reduces a 7×7 assembly transient from 320 s to 4.2 s.
- **Eigenvalue** (criticality $M\varphi = k^{-1}P\varphi$): power iteration, where each step is a driven diffusion solve with the fission source updated from the previous flux.
- **Coupled multiphysics**: Picard iteration between single-physics solves.

All three are captured by one driver, `core.iterate.fixed_point`, on which the power-iteration solver is built. This is the framework’s central finding: genericity across problem *structure*, achieved by composition rather than by special-casing the core.

Table 1: netflow vs published VERA P6/P7 code solutions [2, 3] (code comparison — P6/P7 have no reference solution).

quantity	netflow	published
peak vol-avg fuel T (°C)	999	1066
coolant exit spread (K)	4.9–6.5	6.6 / 8.7
coolant mean rise (K)	37–39	34–37
k convergence (verification)	≤ 0.1 pcm	analytic

3 Verification

Each edge is checked against its analytic resistance; analytic Jacobians agree with finite differences to $\leq 10^{-5}$. The thermal fuel-pin chain reproduces the textbook radial decomposition (film, gap, clad, fuel) [4] to machine precision. Neutron-diffusion k -eff converges to the analytic bare-slab value as $\mathcal{O}(\Delta^2)$ —9.7, 1.5, 0.4, 0.1 pcm at $N = 20, 50, 100, 200$ cells—with the fundamental mode matching the analytic cosine exactly. Global energy balance on a full assembly closes to six significant figures. The suite is 107 automated tests.

4 Code-to-code comparison against VERA

We match the VERA Watts Bar 17×17 geometry [1] exactly. The benchmark’s Problems 6/7 *define inputs only*—the specification states “no reference solution exists” [1]—so published P6/P7 results are themselves code solutions, and this is honestly a code-to-code comparison, not validation against a reference truth. Compared *at the published definition* (VERA reports *volume-average*, not centerline, fuel temperature), netflow’s peak volume-average fuel temperature is 999 °C against the MC21/CTF value of 1066 °C [2] (−6%), for a single hot pin at matched local peaking (1.918 vs. the published 1.92), using the empirical gap conductance (a controlled comparison showed a geometric helium-gap model over-predicts by ~ 220 K). The coolant mean rise (37 K to 39 K) matches the assembly energy balance (~ 36 K). Subchannel exit-coolant spread is a *calibrated* quantity in any scalar model: for the near-flat Problem 6 power, sweeping the lateral-mixing fraction over 0.05–0.40 gives 4.9 K to 6.5 K, essentially matching the published 6.6 K (MC21/CTF, VERA) and under 8.7 K (MC21/COBRA-IE) [3], with guide tubes appearing as cool bypass channels as in the benchmark. The mixing coefficient is calibrated, not predicted: **netflow** is a transparent pre-design tool, not a replacement for CTF/COBRA-TF.

5 Self-consistent multiphysics

Coupling the neutronics and thermal plugins through Picard iteration, with Doppler feedback ($\Sigma_a \propto \sqrt{T}$), *com-*

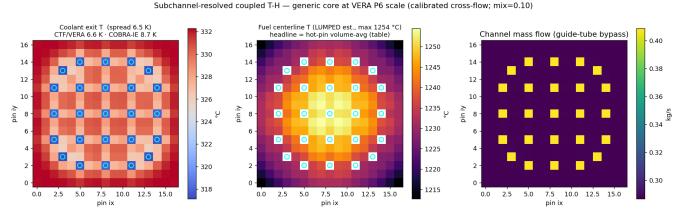


Figure 3: Pin-resolved subchannel coupled thermal-hydraulics (17×17). Left: coolant exit temperature—guide tubes (cyan) form cool spots, matching the VERA pattern. Right: guide-tube bypass flow.

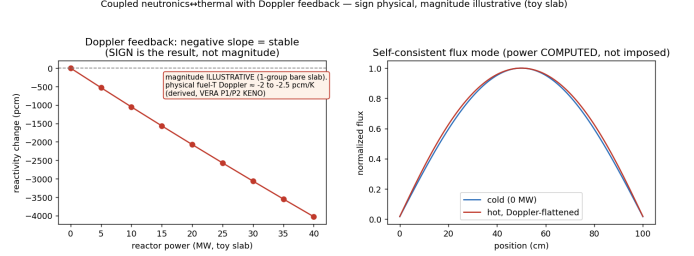


Figure 4: Doppler-coupled neutronics/thermal. Left: reactivity vs. power (negative slope = stable) — the *sign* is the result; the toy-slab magnitude is illustrative (physical ≈ -2 to -2.5 pcm/K). Right: the self-consistent flux mode flattens at power. Power is computed, not imposed.

putes the power distribution rather than imposing it—closing the central caveat of the temperature comparisons. The coupled solve exhibits a *negative* fuel-temperature reactivity coefficient (Fig. 4)—the sign underlying inherent reactor stability—and the flux Doppler-flattens at power. We report this as a mechanism/sign demonstration only: it runs on a one-group bare slab with illustrative cross-sections, so its magnitude is not a physical prediction. The physical reference for fresh 3.10 wt% UO_2 , -2 to -2.5 pcm/K, is itself derived from the spec’s reference KENO-VI eigenvalues for Problems 1–2 [1].

6 Performance and scope

Solve time is linear in node count until sparse-factorization fill-in dominates near 4×10^5 unknowns (~ 190 s for a 1/8-core slice); Newton iteration count is invariant with size. The identified limits—gap-conductance fidelity, subchannel momentum closure, and the $\sim 10^6$ -node factorization wall—are documented as deliberate future work, not hidden. **netflow** targets fast, transparent pre-design scoping accurate to tens of °C, cross-checked against published benchmark code solutions, on a single core that also solves pipe networks and criticality problems. netflow’s node/edge/flux abstraction is the acausal across/through-variable connector of Modelica [7] and ModelingToolkit [8]; it is a minimal, transparent, pure-Python re-derivation of that mature paradigm, not a new one nor a replacement for those tools.

References

- [1] A. T. Godfrey, *VERA Core Physics Benchmark Progression Problem Specifications*, CASL-U-2012-0131-004, Oak Ridge National Laboratory, 2014.
- [2] D. J. Kelly III *et al.*, “MC21/CTF and VERA multiphysics solutions to VERA core physics benchmark progression problems 6 and 7,” *Nucl. Eng. Technol.* **49** (2017) 1326–1338.
- [3] B. N. Aviles *et al.*, “MC21/COBRA-IE and VERA-CS multiphysics solutions to VERA core physics benchmark problem #6,” *Prog. Nucl. Energy* **101** (2017) 338–351.
- [4] N. E. Todreas and M. S. Kazimi, *Nuclear Systems I: Thermal Hydraulic Fundamentals*, 2nd ed., CRC Press, 2012.
- [5] P. Virtanen *et al.*, “SciPy 1.0: fundamental algorithms for scientific computing in Python,” *Nature Methods* **17** (2020) 261–272.
- [6] I. H. Bell *et al.*, “Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp,” *Ind. Eng. Chem. Res.* **53** (2014) 2498–2508.
- [7] Modelica Association, *Modelica Language Specification*; P. Fritzson, *Principles of Object-Oriented Modeling and Simulation with Modelica*, 2nd ed., Wiley, 2014.
- [8] Y. Ma *et al.*, “ModelingToolkit: A Composable Graph Transformation System for Equation-Based Modeling,” arXiv:2103.05244 (2021).